Scalar contact normalization by finite-field avoidance

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Abstract

We construct a scalar contact of numerator degree twenty and denominator degree forty that preserves the exact arithmetic conditions of the specified rational source. A residue avoidance condition ensures a uniform odd valuation at a fixed prime. The required prime exponent is 137731 in the numerator. The construction uses the existing coordinates and finite algebra generators. A general normalization lemma explains the method, and exact finite computations certify all fixed residue claims.

1 A general normalization lemma

Lemma 1. *Let $f \in \mathbb{Z}[X]$ have even degree $d > 0$, and let p be an odd prime not dividing its leading coefficient. Suppose f has no zero modulo p, and that $f(x) \geq m > 0$ for every real x. Choose $a \in \mathbb{F}_p^\times$ such that $-a^{-1} \notin f(\mathbb{F}_p)$. Let C be a positive integer congruent to a modulo p, let L be a positive odd integer, and let $c \in \mathbb{Z}$ be nonzero and prime to p. Assume $C^2 m > c^2 p^L$. Then*

$$R_f(X) = \frac{c^2 p^L f(X)}{(1 + Cf(X))^2}$$

has the following properties for every rational X:

$$0 < R_f(X) < 1, \qquad v_p(R_f(X)) = \begin{cases} L, & v_p(X) \geq 0, \\ L - dv_p(X), & v_p(X) < 0. \end{cases}$$

In particular its p-valuation is odd and at least L. Its numerator and denominator, in lowest terms over $\mathbb{Q}[X]$, have degrees d and $2d$.

Proof. The real denominator is larger than $C^2 f(X)^2$, so

$$0 < R_f < \frac{c^2 p^L}{C^2 f(X)} \leq \frac{c^2 p^L}{C^2 m} < 1.$$

For a p-integral input, both $f(X)$ and $1+Cf(X)$ are p-adic units by their residue assumptions. For an input of valuation $n < 0$, the leading term of $f(X)$ uniquely dominates the other terms, giving $v_p(f(X)) = dn$. Since C is a unit, $v_p(1 + Cf(X)) = dn$. The two formulas follow, and their parity follows from the even degree. Finally $\gcd(f, 1 + Cf) = 1$ in $\mathbb{Q}[X]$. □

The missing residue $-a^{-1}$ is often available without any arithmetic conjecture. For example, if $f(X) = g(X^e)$ modulo p and $\gcd(e, p-1) = q > 1$, then $|f(\mathbb{F}_p)| \leq 1 + (p-1)/q$. If this upper bound is smaller than $p - 1$ and f has no zero, there is a missing nonzero residue. The coefficient C can then be made arbitrarily large in the required residue class; additional divisibility conditions coprime to p can be imposed by the Chinese remainder theorem. Thus real normalization and fixed-prime avoidance can be chosen independently.

2 The explicit scalar contact

Define

$$\beta(X) = 216X^4 + 450459, \qquad r(X) = \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1).$$

Put

$$\ell = 76249, \quad L = 137731, \quad C = 17(676\ell^{68862}+1), \quad R(X) = \frac{102^2\ell^L r(X)}{(1+Cr(X))^2}. \tag{1}$$

Lemma 2. *The integer ℓ is prime. Both $r(X)$ and $1+17r(X)$ have no zero in $\mathbb{F}_\ell$.*

Proof. Trial division through $\lfloor\sqrt{\ell}\rfloor = 276$ proves primality. For a zero of r, one of the five values of β would be $0, 1, 2401, -343/32, 1/2$. The corresponding values of $X^4 = (\beta-450459)/216$ modulo ℓ are

$$32862, \quad 32509, \quad 24048, \quad 904, \quad 70810.$$

Their powers with exponent $(\ell-1)/4$ are respectively $76248, 76248, 76248, 76248, 40177$. None is zero or one, so none is a fourth power.

For the second assertion the exact certificate supplied with this paper checks

$$\gcd(1+17r(X),\ X^{76249}-X) = 1 \quad \text{in } \mathbb{F}_{76249}[X].$$

It records the coefficients of the degree-twenty first polynomial, the remainder of $X^{76249}-X$ modulo it, and a Bézout identity with constant term one. Modular repeated squaring verifies the remainder, and multiplication verifies the Bézout identity. Since $X^{76249}-X$ is the product of the distinct linear polynomials over this field, this proves the assertion. An independent complete residue enumeration gives the same result: X^4 has 19063 values, and $r(X)$ has 16859 values, none equal to -17^{-1}. □

Lemma 3. *For every integer X, $r(X)$ is a square in each of $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, with*

$$v_2(r) = v_3(r) = 2, \qquad v_{17}(r) \in \{0, 2\}.$$

Proof. One has $\beta \equiv 3 \pmod 8$ and $\beta \equiv 18 \pmod{27}$. The two even factors $\beta-1, \beta-2401$ have valuation one, and their odd parts differ by $1200 \equiv 0 \pmod 8$. Their odd-part product is therefore one modulo eight. The three other factors have product $3\cdot 7\cdot 5 \equiv 1 \pmod 8$. Thus $v_2(r) = 2$ and $r/4 \equiv 1 \pmod 8$. At three only β is divisible by three, with valuation two, and

$$r/9 \equiv 2\cdot 2\cdot 2\cdot 1\cdot 2 \equiv 1 \pmod 3.$$

At seventeen the five possible fourth powers and their images are

X^4	0	1	4	13	16
β	10	5	7	13	15
$r(X)$	0	15	2	13	2.

The nonzero entries in the last row are squares. In the first column $17 \mid X$ and

$$32\beta + 343 = 6912X^4 + 17^2\cdot 49879, \qquad 49879 \equiv 1 \pmod{17}.$$

This factor has valuation two, and the other four factors have unit product $10\cdot 9\cdot 6\cdot 2 \equiv 9 \pmod{17}$. The usual local square criteria now give the claim. □

Theorem 4. *The rational function in (1) has the following properties.*

1. *Its numerator and denominator in lowest polynomial terms have degrees twenty and forty, respectively.*

2. *For all* $X \in \mathbb{Q}$, $0 < R(X) < 1$ *and*

$$v_\ell(R(X)) = \begin{cases} 137731, & v_\ell(X) \geq 0, \\ 137731 - 20v_\ell(X), & v_\ell(X) < 0. \end{cases}$$

3. *If* $p > 76176$, $p \neq \ell$, *and* $v_p(r(X)) > 0$, *then* $v_p(R(X)) = v_p(r(X))$.

4. *If* $X \in \mathbb{Z}$, *the reduced denominator of* $R(X)$ *is a square, and* $R(X)$ *is a square in* $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, *with*

$$v_2(R) = v_3(R) = 4, \qquad v_{17}(R) \in \{2, 4\}.$$

Proof. Since $\beta \geq 450459$ on $\mathbb{R}$, all five factors of r are positive and increasing as functions of β. Consequently

$$r(X) \geq r(0) = 11807125278374054049899270780 52.$$

The exact integer inequality

$$36\ell^7 < 676^2 r(0)$$

implies $C^2 r(0) > 17^2 \cdot 676^2 \ell^{137724} r(0) > 102^2 \ell^{137731}$. Now apply Lemma 1 with $f = r$, $d = 20$, $a = 17$, and $c = 102$, using Lemma 2. This gives parts (1) and (2).

For part (3), C is an integer, so $v_p(Cr) > 0$ and the denominator is a unit. The multiplier $102^2\ell^L$ is a p-adic unit for all primes in the stated range.

For integer input let $D_0 = 1 + Cr(X)$. It is coprime to $r(X)$. It is a unit at two and three because r is divisible by four and nine, a unit at seventeen because $17 \mid C$, and a unit at ℓ by Lemma 2. The only prime factors of $102^2\ell^L$ are these four primes. Therefore the displayed numerator is coprime to D_0, so the reduced denominator is exactly D_0^2. Finally $\ell \equiv 1 \pmod 8$, $\ell \equiv 1 \pmod 3$, and $\ell \equiv 4 \pmod{17}$, making ℓ a square in the three completions. Lemma 3 and the square multiplier 102^2 give the stated squareclasses and depths. □

3 Integration into the rational source

The scalar extraction argument of the 29-coordinate source requires a positive rational contact below one, an odd ℓ-valuation at least 137731, preservation of consumed positive contact valuations away from ℓ, and, for integer inputs, a square denominator and the preparation conditions at two, three and seventeen. Theorem 4 supplies these same hypotheses. The finite support can change; the completeness theorem permits preparation at any fixed finite support after fixing the input. Consequently the replacement preserves the source's soundness and completeness under exactly its existing arithmetic premises. No coordinate or finite algebra generator is added.

The final polynomial's degree and coefficient bounds must be obtained from its revised expression graph. The degree-forty denominator and the smaller fixed prime exponent are exact improvements to the scalar source; they do not alone certify a numerical global bound for the eliminated polynomial.

4 A compact residue certificate and reproduction

The root-avoidance calculation also has a small certificate in the variable $U = X^4$. Define four polynomials over $\mathbb{F}_{76249}$ by their coefficient vectors, in ascending order of powers of U:

$$\begin{array}{c|l} F_0 & (2124, 22608, 14921, 12900, 56669, 30599) \\ G_0 & (69700, 50246, 7543, 69483, 32972) \\ A_0 & (32273, 39191, 5614, 60392, 71693) \\ B_0 & (21082, 70119, 44908, 68100, 13136). \end{array}$$

Expansion gives $1+17r(X) = F_0(X^4)$. Repeated squaring and polynomial multiplication give the exact identities

$$U^{19062} - 1 \equiv G_0(U) \pmod{F_0(U)}, \qquad A_0F_0 + UB_0G_0 = 1.$$

If $F_0(x^4) = 0$ for $x \in \mathbb{F}_{76249}$, then $x \neq 0$ since $F_0(0) = 2124$. The element $u = x^4$ satisfies $u^{19062} = 1$, so the first identity gives $G_0(u) = 0$. The second identity then gives $0 = 1$. This supplies the root-avoidance argument with only degree-five arithmetic and one modular exponentiation. The direct residue enumeration is a separate independent verification.

The checker is

```
python check_scalar_normalization.py
```

It uses integer arithmetic and polynomial arithmetic over the prime field, with no floating-point tests. Its output file is

`scalar-normalization-certificate.json`.

This includes the complete modular Bézout certificate. The direct enumeration is independent of the modular polynomial Euclidean algorithm. The valuation formulas for all rational inputs are proved in Lemma 1; no bounded scan is substituted for that argument.